

BERT Variants for Depression Screening with Typed and Transcribed Responses

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ABSTRACT

Early treatment of depression has been proven to be very effective. However, diagnosing depression remains a challenge with current diagnostic tools. Research has begun exploring digital and mobile modalities for depression screening. In our research, we focus on screening for moderate depression with the typed and transcribed responses in the StudentSADD (Student Suicidal Ideation and Depression Detection) dataset. Our modeling approach involves comparing the performance of five BERT (Bidirectional Encoder Representations from Transformers) variants with and without fine-tuning layers. Our results indicate that typed responses are more useful than transcribed responses when screening for depression. In particular, the less computationally expensive BERT variants with both fine-tuning layers performed best for the typed responses. The highest classifier balanced accuracy was 0.63. Our research can help inform the future development of essential digital mental illness screening models.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks; Supervised learning by classification; Natural language processing**; • **Applied computing** → **Health informatics; Psychology**; • **Human-centered computing** → Ubiquitous and mobile devices.

KEYWORDS

digital health; digital biomarkers; transfer learning; text classification

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1 INTRODUCTION

Depression is a mental illness that affected 322 million people globally in 2017 [40]. It is among the top three leading causes of disability worldwide [40], as it affects both physical and role functioning capabilities [16]. Depression has also been cited as the most frequent mental illness associated with suicides [2]. The early recognition and treatment of depression can greatly reduce its health, financial, and social burdens [11].

However, screening for depression remains a challenge. Currently, mental illnesses are primarily screened for using surveys [23]. Research has unfortunately found that people with depression may not recognize their depressive symptoms [10] and are less likely to seek help for depression during the early stages [3]. Mental illness stigma [1, 29] likely also contributes to the low prevalence of early depression screening.

To improve screening rates, researchers are now applying machine learning to data collected from mobile devices in order to develop mental illness screening models. In particular, screening modalities include smartphone logs [4, 26, 28, 31, 37, 38], text messages [25, 30], mobile voice recordings [13, 27, 32], environmental audio [9], and social media [6].

Bidirectional Encoder Representations from Transformers (BERT) [8] pretrain deep bidirectional representations from an unlabeled text by fusing the left and the right context across all layers. This is particularly useful for tasks such as depression screening as context matters greatly. BERT [8] classifiers have been successfully used to classify clinical interview transcripts [34]. Newer variants of BERT have had even more success in classifying mental illnesses with social media [18]. While recent research [24] compared the ability of BERT variants to screen for depression with clinical interview transcripts, no such study has been conducted on the transcripts of mobile voice recordings.

In this research, we thus use BERT classifiers to screen for depression with mobile voice transcripts. In addition to transcripts, we also assess the depression screening ability of typed responses, an underutilized modality in this domain. These text modalities are both included within the StudentSADD (Student Suicidal Ideation and Depression Detection) dataset [27], which was collected from college students after the start of the COVID-19 pandemic to aid the

development of depression screening technologies. We apply five different BERT variants to classify the two types of text. Further, we determine whether fine-tuning layers improves the classification ability of these five variants. Our contributions thus include comparing the depression screening performance of:

- (1) Two different types of text collected from a mobile app,
- (2) Five different pretrained BERT model types, and
- (3) Three different combinations of architectural layers.

2 DATA: TYPED AND TRANSCRIBED TEXT

In this research, we use data from the StudentSADD dataset [27] which was collected from August 2020 to January 2021. StudentSADD contains typed and spoken responses to prompts from over 300 college students, collected via a mobile and web application. Specifically, students were asked to describe their favorite place in 2000 characters or less by typing a response and describe a good friend in 30 seconds or less by speaking into a microphone [27]. The voice recordings were transcribed using Speech Recognition [41].

Prior to the prompts, participants were administered the Patient Health Questionnaire-9 (PHQ-9) [14] to provide depression labels for the data [27]. Popular in primary care settings [23], the PHQ-9 is a nine-question survey tool used to screen for depression. Each PHQ-9 question has a scale from 0 – 3 and relates to the DSM-IV criteria for diagnosing depression [14]. The PHQ-9 score ranges from 0 – 27, with 10 being the common cutoff for moderate depression [14]. As such, we consider participants with a score of at least 10 to screen positive for depression.

StudentSADD [27] consists of text replies from 298 students, 47% of whom screened positive for depression. Likewise, the corpus also consists of transcripts from 110 of the students, 40% of whom screened positive for depression. The high PHQ-9 scores are likely due to the high rates of mental illnesses documented in student populations [4], self-selection bias [9], and the collection occurring within the first year of the COVID-19 pandemic [27].

3 DEEP LEARNING METHODOLOGY

The goal of our research is to explore the depression screening capabilities of popular deep learning models pretrained for text comprehension on large text corpuses, namely BERT and its variants, with mobile transcript and text replies. These pretrained language representation models can be paired with a classifier to be further trained on labeled data.

3.1 BERT Variants

BERT. Utilizing bidirectional architectures, BERT tackles a variety of natural language processing tasks [8]. BERT is pretrained on two text corpuses, BooksCorpus and English Wikipedia, using Masked Language Modeling (MLM) and Next Sentence Prediction (NSP).

RoBERTa. The Robustly Optimized BERT Pretraining Approach (RoBERTa) is a variation of BERT known for better performance [17], including for mental health applications [18]. To improve performance, RoBERTa notably removes the NSP objective, utilizes dynamic MLM, and trains on longer text sequences from three additional corpuses [17].

ALBERT. A Lite BERT, ALBERT is a variation of BERT that addresses the computational problems that one faces while pre-training a large model size [15]. ALBERT achieves this by utilizing two parameter reduction techniques that lower memory consumption and increase training speed [15].

DistilBERT. The distilled version of BERT, DistilBERT, adopts the BERT model that allows the functionality of BERT while requiring less computational power [22]. DistilBERT distills knowledge during the pretraining phase, reducing the size while maintaining 97% of BERT’s language understanding and being 60% faster [22].

BERTweet. BERTweet is a publicly available largescale pretrained model trained using English tweets [19]. BERTweet utilizes the same architecture as BERT and is trained using the RoBERTa pretraining procedure [19].

3.2 Fine-Tuning Layers

To improve the classification ability of the BERT variations, we implement two additional architectural layers to augment each of the five base uncased BERT models. The first fine-tuning layer is the Long Short-Term Memory (LSTM) [12], known to be advantageous for longer sequences. We also add an extra self-attention [36] fine-tuning layer to improve the identification of relevant information within text sequences. Thus, our three architectures include the base BERT model (B), the addition of an LSTM layer to the base BERT model (BL), and the addition of both an LSTM and a self-attention layer to the base BERT model (BLA).

3.3 Experimental Evaluation

Implementation details. We use the stratified test set defined in the StudentSADD paper [27] which consists of approximately a quarter of the data. We upsample the training set to remedy the class imbalance. All of our classifiers employ a cross-entropy function and an Adam optimizer.

Based on the results of our parameter search, we use a learning rate of $5e^{-5}$ and batch size of 1 for typed responses as well as a learning rate of $2e^{-6}$ and batch size of 4 for transcribed responses. Further, all models use a step size of $2e^{-8}$ and 128 maximum number of tokens. We train models for 20 epochs and employ early stopping as needed. We repeat each experiment 20 times for robustness.

Evaluation metrics. To evaluate the performance of our models, we use three common metrics: balanced accuracy, sensitivity, and specificity. Sensitivity measures the model’s ability to correctly classify a depressed individual as depressed whereas specificity measures its ability to correctly classify a not depressed individual as not depressed. These are inversely proportional metrics important for assessing the usefulness of diagnostic models. Meanwhile, balanced accuracy is the average of sensitivity and specificity, making it an ideal metric to assess the overall performance of the classifiers. To mitigate the impact of randomly initialized weights on the results, we report on average and standard deviation of the 10 out of the 20 model runs with the highest balanced accuracies.

3.4 Software and Research Updates

The BERT models we utilized in this research are from Hugging Face [39]. We will also post updates about the StudentSADD dataset and our future research on our project website: <https://emutivo.wpi.edu>.

4 RESULTS

The balanced accuracies for the 15 models are displayed in Fig. 1. The typed responses are more predictive of depression than the transcribed responses. For the typed responses, the best model was DistilBERT with both fine-tuning layers; it achieved a balanced accuracy of 0.63 ± 0.04 , sensitivity of 0.46 ± 0.08 , and specificity of 0.66 ± 0.08 . In contrast, the best models with transcripts only achieved a balanced accuracy of 0.56 ± 0.04 . The results of these models for typed and transcribed responses are detailed in Table 1 and Table 2, respectively.

4.1 Typed Responses

BERT variants. As mentioned, DistilBERT with both architectural layers was the most successful on the typed responses. ALBERT also performed well with both architectural layers, achieving an average balanced accuracy only 0.03 below that of the best model. Both of these models are trained similar to BERT but are less computationally expensive. Unexpectedly, RoBERTa and BERTweet, the more complicated models, performed the worst on the typed responses. At times, simpler models can be more effective than unnecessarily complex models given that they are harder to fine-tune due to their complexity. In summary, the typed responses performed better with the less complicated BERT variants, which is promising for the future of mental illness screening technologies.

Architectural layers. Fine-tuning layers were beneficial to the depression screening ability of the models using the typed responses. For each variant, the base model without any fine-tuning layers performed the worst. With the exception of BERT, both fine-tuning layers produced the best results. This improvement in balanced accuracy was most evident for ALBERT which experienced an increase of 0.09 between the least and most architectural layers. Even RoBERTa, which benefited least from the architectural layers, still improved the average balanced accuracy by 0.04. In particular, the addition of both the LSTM and attention layers seems to benefit most BERT variants for typed responses.

4.2 Transcribed Responses

BERT variants. ALBERT and basic BERT, both without fine-tuning layers, performed the best with transcribed responses. However, even these models only yielded an average balanced accuracy that was 0.05 better than BERTweet, the worst performing variant. Overall, no variants were particularly promising, perhaps suggesting that BERT is not an effective approach for transcribed responses.

Architectural layers. Unlike for the typed responses, the fine-tuning layers did not notably improve the balanced accuracies of the models with the transcribed responses. In particular, models with both fine-tuning layers performed the worst for all five BERT variants. While models with just the LSTM layer did perform better than models with no fine-tuning layers for DistilBERT and BERTweet, the improvement was only 0.02 and 0.01, respectively. In particular, the addition of both the LSTM and attention layers seems to benefit none of the BERT for transcribed responses.

5 DISCUSSION & FUTURE WORK

The typed responses were notably more useful at screening for depression than the transcribed responses. As more students shared

the typed responses, the larger quantity could have led to higher results. Yet, the set of participants with transcripts is similar in size to that of the set of participants with clinical interview transcripts in the related literature, which were able to be successfully modeled by BERT [34]. We hypothesize the lack of predictability of StudentSADD transcripts may therefore be due to the diverse dialects of the domestic and international student participants. Further, the mobile audio quality, the selected prompt, and the lack of a clinical context could also have influenced the predictability. Since BERT variants were insufficient for classifying the mobile transcribed responses, future research could experiment with alternate pretrained deep learning models, such as more recent the third-generation Generative Pre-trained Transformer (GPT3) [5] and Text-to-Text Transfer Transformer (T5) model [20].

While research screening for mental illnesses with voice recordings is relatively common [7, 33, 35], typed responses in the form of narratives are much less studied [27]. It was presumed the rich multimodal nature of voice [34] would yield modalities more predictive than a typed response. However, our results indicate that transcribed responses are less predictive than typed responses. Voice recordings were also less likely to be shared by students and pose a privacy risk [21, 27]. As typed responses do not suffer from the aforementioned limitations of the transcribed responses, our research recommends the future use of this underutilized modality in screening technologies.

We were able to improve upon the previously reported results for typed responses [27] by utilizing BERT variants with fine-tuning layers. We acknowledge that our results are still unfortunately insufficient to be used alone to screen for depression. We thus recommend future research focus on incorporating typed responses in multimodal screening models.

6 CONCLUSION

In this research, we evaluate the ability of BERT variants to improve depression screening with two mobile text modalities. Our results indicate that typed responses are more useful than transcribed responses when screening for depression. Using BERT variants, we were able to improve upon the previously reported screening capabilities of the typed responses. In particular, DistilBERT with both fine-tuning layers achieved a balanced accuracy of 0.63, sensitivity of 0.46, and specificity of 0.66.

From our results, we deduce that the computationally less expensive models are better at screening for depression with typed responses. Unlike for the transcribed responses, the fine-tuning layers proved beneficial for all BERT variants using the typed responses. While typed responses are underutilized in mobile health, they are a promising screening modality that could be easily incorporated into multimodal models. Our findings thus provide useful information to guide future digital mental illness screening research.

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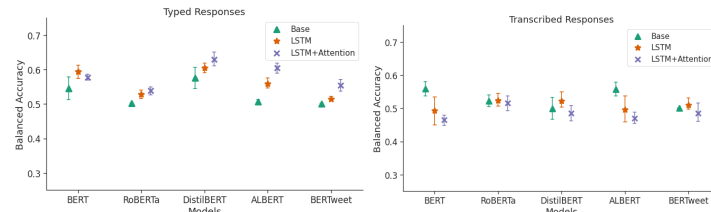


Figure 1: Result comparison of the five BERT variants, with three unique architectures, on the two StudentSADD text modalities.

Table 1: Results of the five BERT variants, with three different architectures, for the StudentSADD typed responses.

Model Architecture	Balanced Accuracy			Sensitivity			Specificity		
	B	BL	BLA	B	BL	BLA	B	BL	BLA
BERT	0.54±0.05	0.59±0.03	0.58±0.01	0.43±0.17	0.49±0.05	0.52±0.09	0.74±0.18	0.75±0.06	0.66±0.09
RoBERTa	0.50±0.00	0.53±0.02	0.54±0.02	0.88±0.30	0.73±0.17	0.42±0.20	0.12±0.28	0.59±0.27	0.68±0.08
ALBERT	0.51±0.01	0.56±0.02	0.60±0.02	0.97±0.08	0.52±0.06	0.59±0.07	0.00±0.00	0.74±0.03	0.70±0.06
DistilBERT	0.58±0.05	0.61±0.02	0.63±0.04	0.45±0.07	0.57±0.18	0.46±0.08	0.10±0.16	0.55±0.25	0.66±0.08
BERTweet	0.50±0.00	0.51±0.01	0.55±0.03	0.69±0.41	0.80±0.23	0.70±0.19	0.66±0.47	0.41±0.32	0.70±0.07

Table 2: Results of the five BERT variants, with three different architectures, for the StudentSADD transcribed responses.

Model Architecture	Balanced Accuracy			Sensitivity			Specificity		
	B	BL	BLA	B	BL	BLA	B	BL	BLA
BERT	0.56±0.04	0.49±0.07	0.46±0.03	0.86±0.04	0.42±0.41	0.65±0.24	0.32±0.04	0.65±0.36	0.36±0.23
RoBERTa	0.52±0.03	0.52±0.03	0.52±0.04	0.85±0.14	0.81±0.27	0.85±0.20	0.39±0.20	0.70±0.27	0.54±0.17
ALBERT	0.56±0.04	0.50±0.07	0.47±0.03	0.76±0.16	0.72±0.33	0.62±0.35	0.36±0.30	0.68±0.36	0.42±0.21
DistilBERT	0.50±0.06	0.52±0.04	0.49±0.05	0.78±0.28	0.52±0.36	0.63±0.34	0.67±0.30	0.61±0.31	0.58±0.14
BERTweet	0.50±0.00	0.51±0.03	0.49±0.05	0.00±0.01	0.86±0.28	0.68±0.29	0.02±0.06	0.44±0.22	0.57±0.17

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